1. The value of sin 90° is

A) 0

B) 1

C) -1

D) 1/2

Answer: B

---

2. The value of cos 0° is

A) 0

B) 1

C) -1

D) 1/2

Answer: B

---

3. The derivative of $x^2$ is

A) x

B) 2x

C) $x^2$

D) 2

Answer: B

---

4. The integration of 1 dx is

A) x

B) $x^2$

C) 1

D) 0

Answer: A

---

5. The value of tan 45° is

A) 0

B) 1

C) -1

D) √3

Answer: B

---

6. The value of $\sin^2\theta + \cos^2\theta$ is

A) 0

B) 1

C) 2

D) -1

Answer: B

---

7. The determinant of a 2×2 matrix is

A) ad + bc

B) ad − bc

C) a + b + c + d

D) abcd

Answer: B

---

8. The slope of a horizontal line is

A) 0

B) 1

C) Infinity

D) -1

Answer: A

---

9. The slope of a vertical line is

A) 0

B) 1

C) Infinity

D) -1

Answer: C

10. The value of $\log_{10}10$ is

A) 0

B) 1

C) 10

D) -1

Answer: B

---

11. The value of $\log_{10}1$ is

A) 0

B) 1

C) 10

D) -1

Answer: A

---

12. The solution of $x^2 = 0$ is

A) $x = 1$

B) $x = 0$

C) $x = -1$

D) No solution

Answer: B

---

13. The number of solutions of a linear equation is

A) One

B) Two

C) Infinite

D) None

Answer: A

---

14. The distance between two points formula involves

A) Addition

B) Subtraction

C) Square root

D) Division

Answer: C

---

15. The formula of circle is

A) $x^2 + y^2 = r^2$

B) $x + y = r$

C) $x^2 - y^2 = r^2$

D) $xy = r^2$

Answer: A

---

16. The value of sin 0° is

A) 0

B) 1

C) -1

D) 1/2

Answer: A

---

17. The value of cos 90° is

A) 0

B) 1

C) -1

D) 1/2

Answer: A

---

18. The derivative of sin x is

A) cos x

B) -cos x

C) sin x

D) -sin x

Answer: A

---

19. The derivative of cos x is

A) sin x

B) -sin x

C) cos x

D) -cos x

Answer: B

---

20. The integration of x dx is

A) $x^2/2$

B) $x^2$

C) 2x

D) x

Answer: A

---

21. The value of tan 0° is

A) 0

B) 1

C) -1

D) Undefined

Answer: A

---

22. The value of tan 90° is

A) 0

B) 1

C) Undefined

D) -1

Answer: C

---

23. The number π is approximately

A) 2.14

B) 3.14

C) 4.14

D) 1.14

Answer: B

---

24. The area of a circle is

A) $\pi r^2$

B) $2\pi r$

C) $r^2$

D) $\pi r$

Answer: A

---

25. The perimeter of circle is

A) $\pi r^2$

B) $2\pi r$

C) $r^2$

D) $\pi r$

Answer: B

---

26. The value of $\sqrt{16}$ is

A) 2

B) 4

C) 8

D) 16

Answer: B

---

27. The value of $\sqrt{25}$ is

A) 5

B) 25

C) 10

D) 15

Answer: A

---

28. The value of $2^3$ is

A) 6

B) 8

C) 9

D) 4

Answer: B

---

29. The value of $3^2$ is

A) 6

B) 9

C) 3

D) 12

Answer: B

---

30. The value of $5^0$ is

A) 0

B) 1

C) 5

D) Undefined

Answer: B

---

31. The formula of $(a+b)^2$ is

A) $a^2 + b^2$

B) $a^2 + 2ab + b^2$

C) $a^2 - b^2$

D) $a^2 - 2ab + b^2$

Answer: B

---

32. The formula of $(a-b)^2$ is

A) $a^2 - 2ab + b^2$

B) $a^2 + 2ab + b^2$

C) $a^2 - b^2$

D) $a^2 + b^2$

Answer: A

---

33. The value of $10^2$ is

A) 10

B) 20

C) 100

D) 200

Answer: C

---

34. The value of 1/0 is

A) 0

B) 1

C) Undefined

D) Infinity

Answer: C

---

35. The graph of $y = x^2$ is

A) Line

B) Circle

C) Parabola

D) Hyperbola

Answer: C

---

36. The solution of $x + 2 = 5$ is

A) 2

B) 3

C) 5

D) 7

Answer: B

---

37. The value of sin 30° is

A) 1/2

B) √3/2

C) 1

D) 0

Answer: A

---

38. The value of cos 60° is

A) 1/2

B) √3/2

C) 1

D) 0

Answer: A

---

39. The value of tan 30° is

A) 1/√3

B) √3

C) 1

D) 0

Answer: A

---

40. The value of tan 60° is

A) 1/√3

B) √3

C) 1

D) 0

Answer: B

41. The number of diagonals in a square is

A) 2

B) 4

C) 1

D) 3

Answer: A

---

42. The sum of angles of triangle is

A) 90°

B) 180°

C) 360°

D) 270°

Answer: B

---

43. The sum of angles of a quadrilateral is

A) 180°

B) 360°

C) 270°

D) 90°

Answer: B

---

44. The probability value lies between

A) 0 and 1

B) 1 and 2

C) -1 and 1

D) 0 and 2

Answer: A

---

45. The probability of impossible event is

A) 0

B) 1

C) 1/2

D) -1

Answer: A

---

46. The probability of certain event is

A) 0

B) 1

C) 1/2

D) -1

Answer: B

---

47. Mean of data is

A) Sum / Number

B) Number / Sum

C) Product

D) Difference

Answer: A

---

48. Median is

A) Middle value

B) Largest value

C) Smallest value

D) Average

Answer: A

---

49. Mode is

A) Most frequent value

B) Least value

C) Average

D) Sum

Answer: A

---

50. The equation of straight line is

A) $y = mx + c$

B) $x^2 + y^2 = r^2$

C) $xy = c$

D) $x + y^2 = c$

Answer: A

---

51. The distance between (0,0) and (3,4) is

A) 5

B) 6

C) 7

D) 4

Answer: A

---

52. The midpoint formula gives

A) Centre point

B) End point

C) Distance

D) Area

Answer: A

---

53. The value of $\log_2 8$ is

A) 2

B) 3

C) 4

D) 1

Answer: B

---

54. The value of $\log_{10} 0.1$ is

A) 1

B) -1

C) 0

D) 10

Answer: B

---

55. The inverse of addition is

A) Multiplication

B) Division

C) Subtraction

D) Power

Answer: C

---

56. The inverse of multiplication is

A) Addition

B) Division

C) Subtraction

D) Square

Answer: B

---

57. The domain of $\sqrt{x}$ is

A) $x \geq 0$

B) $x \leq 0$

C) All real numbers

D) $x > 1$

Answer: A

---

58. The range of sin x is

A) [-1,1]

B) [0,1]

C) $(-\infty,\infty)$

D) $[1,\infty)$

Answer: A

---

59. The value of $e^0$ is

A) 0

B) 1

C) e

D) Undefined

Answer: B

---

60. The derivative of constant is

A) 0

B) 1

C) Constant

D) Undefined

Answer: A